Literature-Implied Answer To A C*-Case Isometry Rigidity Question

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Status

Status: literature-implied answer, partial C*-subcase, with the strengthened conclusion of bijective coarse equivalence.

Source question: Bruno M. Braga, *On Banach algebras of band-dominated operators and their order structure*, arXiv:2003.11567. The relevant questions appear in Section 7, "Open questions", as Problem 7.1 (source TeX label `Prob2`) and the following C*-case Problem 7.2 (source TeX label `ProblemIsometryRig`).

Supporting theorems: Florent P. Baudier, Bruno M. Braga, Ilijas Farah, Ana Khukhro, Alessandro Vignati, and Rufus Willett, *Uniform Roe algebras of uniformly locally finite metric spaces are rigid*, arXiv:2106.11391, proved that $*$-isomorphism of uniform Roe algebras implies coarse equivalence. Alessandro Vignati, *The rigidity problem for uniform Roe algebras*, arXiv:2602.01998, strengthens this as Theorem `thmi:main`: if X and Y are uniformly locally finite metric spaces and $\mathrm{C}^*_u(X)$ and $\mathrm{C}^*_u(Y)$ are $*$-isomorphic, then X and Y are bijectively coarsely equivalent.

Source Question

Problem 7.1 of arXiv:2003.11567 asks whether, for uniformly locally finite metric spaces $(\mathbb{N}, d)$ and $(\mathbb{N}, \partial)$ and strictly convex Banach spaces with 1-symmetric bases, a Banach-space isometry

$$B_u(d, \mathcal{E}) \cong B_u(\partial, \mathcal{F})$$

forces the two metric spaces to be coarsely equivalent. The paper explicitly notes that this is open even in the Hilbert case $E = \ell_2$ with its standard basis. The subsequent Problem 7.2 asks, for uniformly locally finite metric spaces X, Y, whether the following two hypotheses imply coarse equivalence:

$$\mathrm{C}^*_u(X) \text{ and } \mathrm{C}^*_u(Y) \text{ are isometric as Banach spaces,}$$

and

$$\mathrm{C}^*_u(X) \text{ and } \mathrm{C}^*_u(Y) \text{ are isomorphic as ordered Banach spaces.}$$

Identification

Under the standard complex-linear reading of "isometric as Banach spaces", the C*-case has a positive answer. Indeed, let

$$T : \mathrm{C}^*_u(X) \to \mathrm{C}^*_u(Y)$$

be a surjective complex-linear isometry. Kadison's theorem on isometries of C*-algebras says that $u = T(1)$ is a unitary and

$$J(a) = u^* T(a)$$

is a unital Jordan $*$-isomorphism.

The uniform Roe algebra $\mathrm{C}^*_u(Y)$ contains $\mathcal{K}(\ell_2(Y))$ as an essential simple ideal. Therefore $\mathrm{C}^*_u(Y)$ is prime: every nonzero closed ideal intersects $\mathcal{K}(\ell_2(Y))$ nontrivially, hence contains $\mathcal{K}(\ell_2(Y))$, so two nonzero closed ideals cannot have zero product. A standard structure theorem for Jordan $*$-homomorphisms then implies that a Jordan $*$-isomorphism onto a prime C*-algebra is either a $*$-isomorphism or a $*$-anti-isomorphism.

If J is multiplicative, $J : \mathrm{C}^*_u(X) \to \mathrm{C}^*_u(Y)$ is already a $*$-isomorphism. If J is anti-multiplicative, compose it with the transpose anti-automorphism

$$\tau_Y([a_{yz}]) = [a_{zy}]$$

of $\mathrm{C}^*_u(Y)$. This transpose preserves finite propagation and hence preserves the norm closure defining $\mathrm{C}^*_u(Y)$. Thus $\tau_Y \circ J$ is a $*$-isomorphism from $\mathrm{C}^*_u(X)$ onto $\mathrm{C}^*_u(Y)$.

In either case, $\mathrm{C}^*_u(X)$ and $\mathrm{C}^*_u(Y)$ are $*$-isomorphic. Theorem `thmi:main` of Vignati's 2026 paper then gives that X and Y are bijectively coarsely equivalent. The earlier theorem of Baudier–Braga–Farah–Khukhro–Vignati–Willett already gave coarse equivalence; the later theorem upgrades the conclusion to the bijective form.

Conclusion And Scope

The C*-subcase of Problem 7.1 has a positive literature-implied answer:

$$\begin{aligned} \mathrm{C}^*_u(X) \cong \mathrm{C}^*_u(Y) \quad & \text{isometrically as complex Banach spaces} \\ \Longrightarrow \quad & X \text{ and } Y \text{ are bijectively coarsely equivalent.} \end{aligned}$$

Consequently, Problem 7.2 has a positive answer, and its additional ordered Banach space isomorphism hypothesis is not needed in the C*-case.

This does not settle the full Problem 7.1 for arbitrary strictly convex Banach spaces with 1-symmetric bases. The argument uses Kadison's C*-algebraic isometry theorem and the later unconditional rigidity theorem for uniform Roe C*-algebras.

Search Evidence

The relation is agent-identified rather than explicitly advertised by the supporting papers as an answer to the exact source problem. The cheap run indexes were searched for arXiv:2003.11567, arXiv:2602.01998, the paper titles, "band-dominated operators", "uniform Roe", "isometric as Banach spaces", "bijective coarse equivalence", and "order structure". The prior version of this packet used arXiv:2106.11391 for coarse equivalence; the present update adds arXiv:2602.01998, whose abstract and Theorem `thmi:main` provide the strengthened bijective rigidity conclusion after the Kadison/Jordan reduction.

References

[1] B. M. Braga, *On Banach algebras of band-dominated operators and their order structure*, J. Funct. Anal. 280 (2021), no. 9, Paper No. 108958; arXiv:2003.11567.

[2] F. P. Baudier, B. M. Braga, I. Farah, A. Khukhro, A. Vignati, and R. Willett, *Uniform Roe algebras of uniformly locally finite metric spaces are rigid*, Invent. Math. 230 (2022), 1071–1100; arXiv:2106.11391.

[3] A. Vignati, *The rigidity problem for uniform Roe algebras*, arXiv:2602.01998, 2026.

[4] R. V. Kadison, *Isometries of operator algebras*, Ann. of Math. (2) 54 (1951), 325–338.